Magnetic field reconstruction with differential phase contrast

MAT SCI 465: Advanced Electron Microscopy and Diffraction

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March 21, 2026

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GitHub Repository: https://github.com/pravan007/MATSCI465_final_project

Abstract

Differential phase contrast (DPC) has emerged as a useful technique in 4-dimensional scanning transmission electron microscopy (4D-STEM) for detecting electromagnetic fields present within a sample. At its most elementary, the technique involves the measurement of beam deflection due to the Lorentz force imposed on the electron beam by the sample's fields. This article details the development of a pipeline that calculates this beam shift, and thus the magnetic fields, in two magnetically patterned Permalloy samples (one circular and one rectangular). First, the beam shifts are calculated via two methods, center-of-mass (CoM) and edge cross-correlation (ECC), which respectively track movement of the center and edges of the elliptical beam in diffraction space. From there, a descan correction is implemented in real space, and then phase gradient and magnetic induction maps are computed directly from the beam shifts. The field behavior of both samples agrees with expectations from the literature, with a counterclockwise vortex in the disk and Landau patterning in the rectangle. The ECC method produced cleaner maps, indicating a potentially useful alternative to the CoM method for materials with complex domain structures. Future work includes potentially moving the descan correction step earlier in the pipeline, and adding external fields to the sample to verify that the resulting additional beam shifts are recognized.

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1 Introduction and Background

In the realm of nanoscale magnetic materials, it is often important to know the magnitude and direction of the electromagnetic fields present within a material sample. This is because of the co-dependence of the material's microstructural properties and its responses to external fields, both of which influence its applicability in memory, logic, and data storage devices [1]. Differential phase contrast (DPC) has emerged as a technique in 4-dimensional scanning transmission electron microscopy (4D-STEM) capable of detecting these electromagnetic fields through capturing their interaction with the microscope's electron beam [2]. Most fundamentally, DPC imaging of magnetic samples requires consideration of the phase shift (caused by the quantum-mechanical Aharanov-Bohm effect) exerted by the fields on the transmitted electrons. However, these effects are also easily understood purely through the classical Lorentz force

$$\vec{F} = q(\vec{v} \times \vec{B})$$

in which the electrons are deflected in a direction normal to both the beam and the magnetic field [3]. By calculating the direction and magnitude of these beam shifts due to the fields in a sample, one can determine the phase gradient and therefore the field.

The goal of this project is to develop a pipeline that takes experimental 4D-DPC data, calculates the beam shifts, and produces a map of the magnetic induction. Previous work has been done both on calculating beam shifts from raw data and on reconstructing magnetic fields from phase gradients, but unifying these different approaches into a single magnetic DPC-focused pipeline would be useful.

2 Methods

2.1 Dataset description

Two datasets are used in this project: one for a 2-micron Permalloy disk, and one for a 3×1.5 micron Permalloy rectangle (see Figure 1). Permalloy is a nickel-iron alloy often used in magnetic cores due to its high magnetic permeability. It is chosen here also because its magnetic patterning is well-studied enough that the expected fields in both sample shapes are known (see Section 3.4). Both samples have a thickness of 18 nm with a 2-nm platinum coating. Below a thickness of about 35 nm, Permalloy exhibits Bloch-type domain walls, in which the magnetization rotates from one magnetic domain to the next in the plane of the domain wall. On the other hand, Néel-type domain walls exhibit out-of-plane magnetization rotations, leading to magnetic effects that DPC cannot capture [4]. This is one of the reasons DPC only works under a certain thickness limit for magnetic materials.

Each dataset is about 2 GB in size, and had 256×256 pixel diffraction images. The real-space image dimensions are 128×128 for the disk and 157×99 for the rectangle. Data was collected by a Spectra microscope with a pixelated CCD detector, meaning that beam shifts are available pixel by pixel. This is contrasted with segmented detectors, which calculate beam shifts through the difference in beam areas in each segment. Segmented detectors produce much smaller datasets and perform deflection angle calculations more quickly; however, pixelated detectors are easier to integrate with other microscope setups and workflows, and have been shown to yield better images for magnetic samples [3].

2.2 Computational approach

Beam shifts were calculated via two methods: center of mass (CoM) and edge cross-correlation (ECC). In the CoM method, the center of each beam is obtained and then tracked over the diffraction space, relative to a reference point. In the ECC method, the beam's edges are tracked rather than its center. ECC is

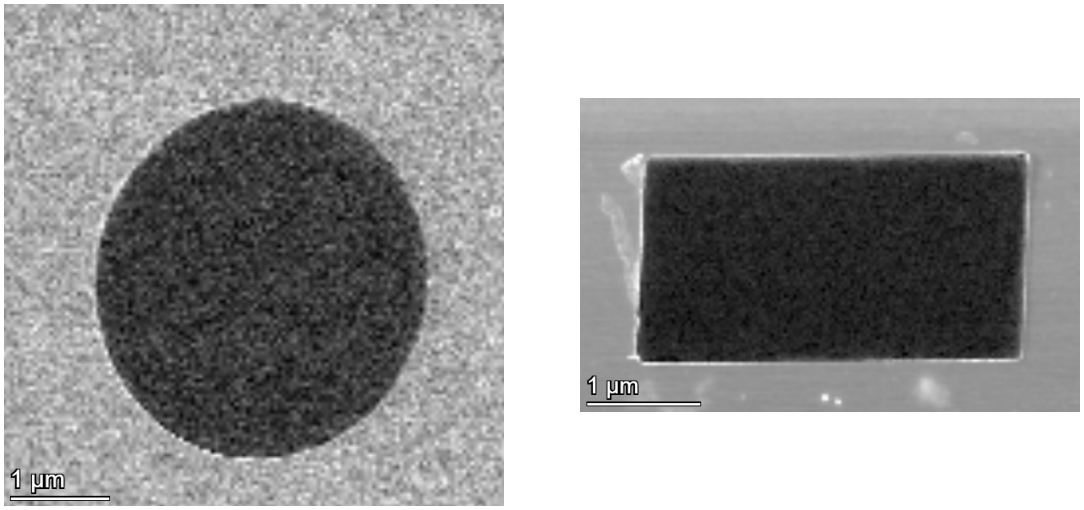

Figure 1: HAADF images of the Permalloy disk and rectangle.

more robust against instances of non-uniform diffraction beams, unwanted diffraction contrast, or dynamical scattering artifacts that may lead to inaccuracies when using the CoM method.

All code was done in Python. Analysis uses a custom package, `dpc4dstem`, developed by Daniel Durham in 2023 and with slight modifications by the author. This package includes some key functions for data handling and visualizing, particularly for the edge cross-correlation method. It also contains code for generating a movie tracking the beam shifts over a single axis in the real-space image (a few such movies are included in the GitHub). Lastly, visualization of the final magnetic induction is done with the PyLorentz package [5], which is primarily used for reconstructing phase shifts in Lorentz transmission electron microscopy (LTEM) data.

3 Results and Discussion

3.1 Center-of-mass disk tracking

After initializing and preparing the 4D dataset, the x- and y-coordinates of the center of the diffraction beam are obtained via the formulas

$$\mathrm{CoM}_x = \frac{\langle I \cdot x \rangle}{\langle I \rangle}$$

$$\mathrm{CoM}_y = \frac{\langle I \cdot y \rangle}{\langle I \rangle}$$

with I being the image intensity values and the angled brackets representing means. Figure 2 shows this calculation for both a raw and filtered diffraction image.

The center of mass is calculated for every pixel, and the beam shift for a given pixel is simply the difference between the CoM at that pixel and the CoM at a background reference point, which is taken to be (0,0). From this, one can plot separately the x- and y-shifts as heatmaps (Figure 3).

3.2 Edge cross-correlation disk tracking

Similar to CoM, the ECC method measures the shifts of the diffraction beam relative to a reference point. As part of the process, the CoM is calculated via the same formula as above; however, it is only used

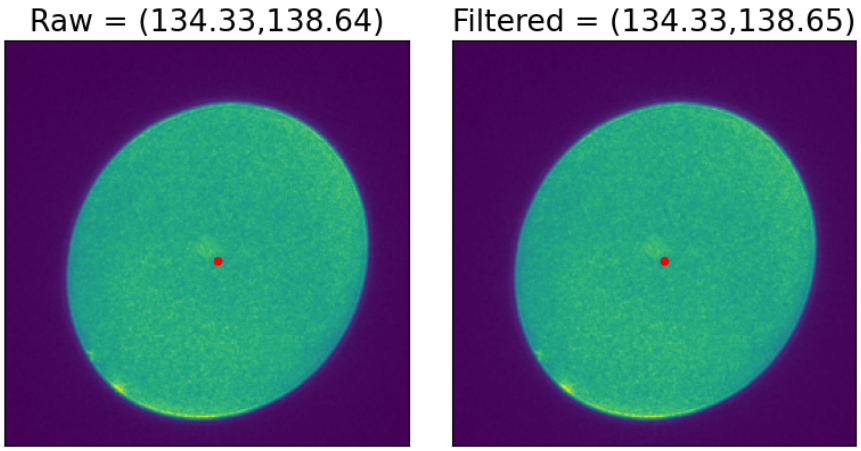

Figure 2: Example CoM calculation for a raw and filtered diffraction image. In this and all other tested cases, the difference between the two is in hundredths of a pixel and therefore negligible.

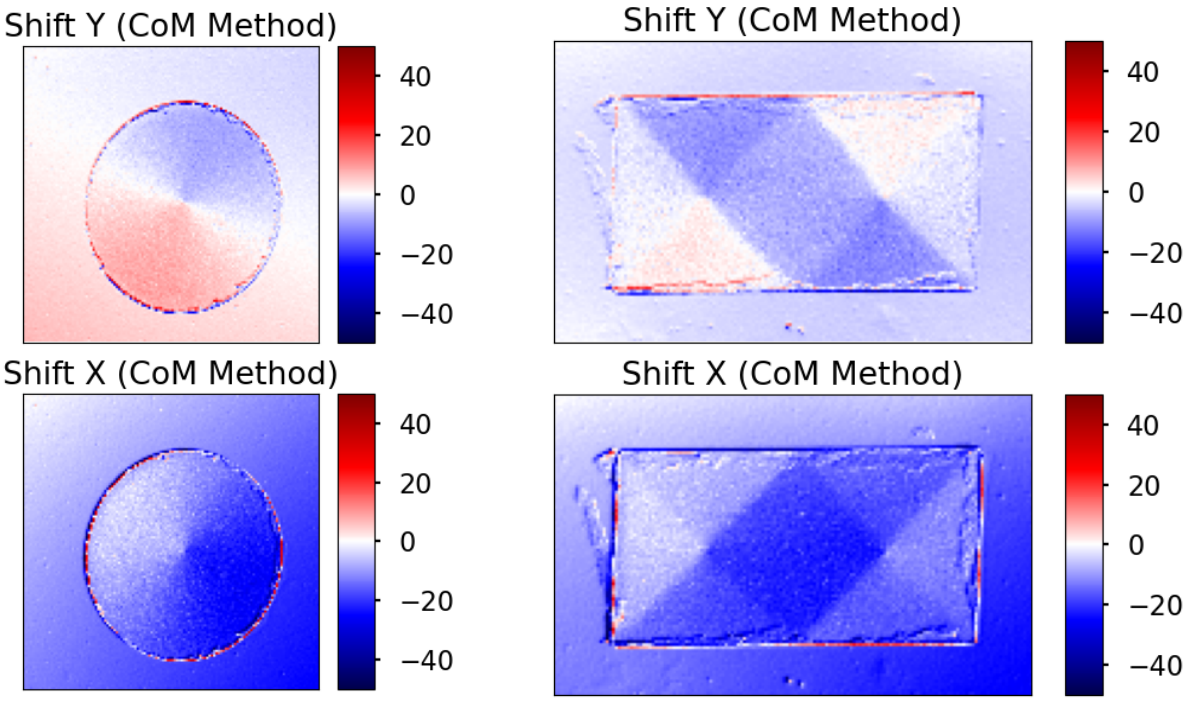

Figure 3: Beam shifts for the disk and rectangle calculated via the CoM method.

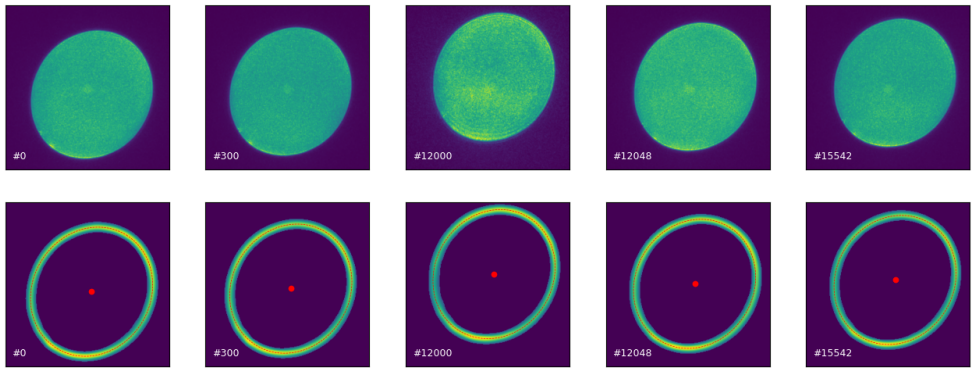

Figure 4: Example of ECC ellipse fits for various pixels.

to generate an elliptical fit for the beam. This fit is initially expressed in the standard form

$$Ax^2 + Bxy + Cy^2 = 1$$

but is then converted (via a py4DSTEM function) into parametric coordinates

$$x = x_0 + A\cos\theta\cos t + B\sin\theta\sin t$$

$$y = y_0 + A\sin\theta\cos t + B\cos\theta\sin t$$

where a and b are the major and minor radii and θ is the tilt angle. This representation is reminiscent of the "elliptical Hough transform," which detects ellipses based on these five parameters (center coordinates, radii, and tilt). As shown in Figure 4, this method tracks the ellipse fairly well over shifts in diffraction space.

3.3 Phase gradient and magnetic induction calculation

From the raw beam shifts as displayed in Figures 3 and 5, one can partially see the presence of at least some domains; i.e. there are discrete patterns where shifts in one direction are particularly strong. However, further processing is necessary for these patterns to be fully interpretable. Firstly, the x maps in both samples display a prevailing leftward shift, including in the background. In STEM, electron beams scanned over a wide area tend to deviate from the normal optical axis, especially near the edges of the image. Accounting for this microscope (non-magnetic) effect is called "descan correction," and it is implemented here by masking out the sample, performing a planar fit on the background, and subtracting this fit from the whole image. Figure 6 shows an example of the masks applied to a beam shift map, and Figures 7 and 8 show the descan-corrected beam shifts for CoM and ECC, respectively.

Once descan correction is complete, the phase gradients can be easily determined through the relationship

$$\boldsymbol{\beta} = -\frac{\lambda}{2\pi}\boldsymbol{\nabla}_\perp\phi$$

where $\boldsymbol{\beta}$ is the deflection angle and λ is the de Broglie wavelength of the setup (here, the electrons have an acceleration voltage of 200 kV, making $\lambda = 2.51$ pm) [6]. Phase gradient maps are shown in Figure 9.

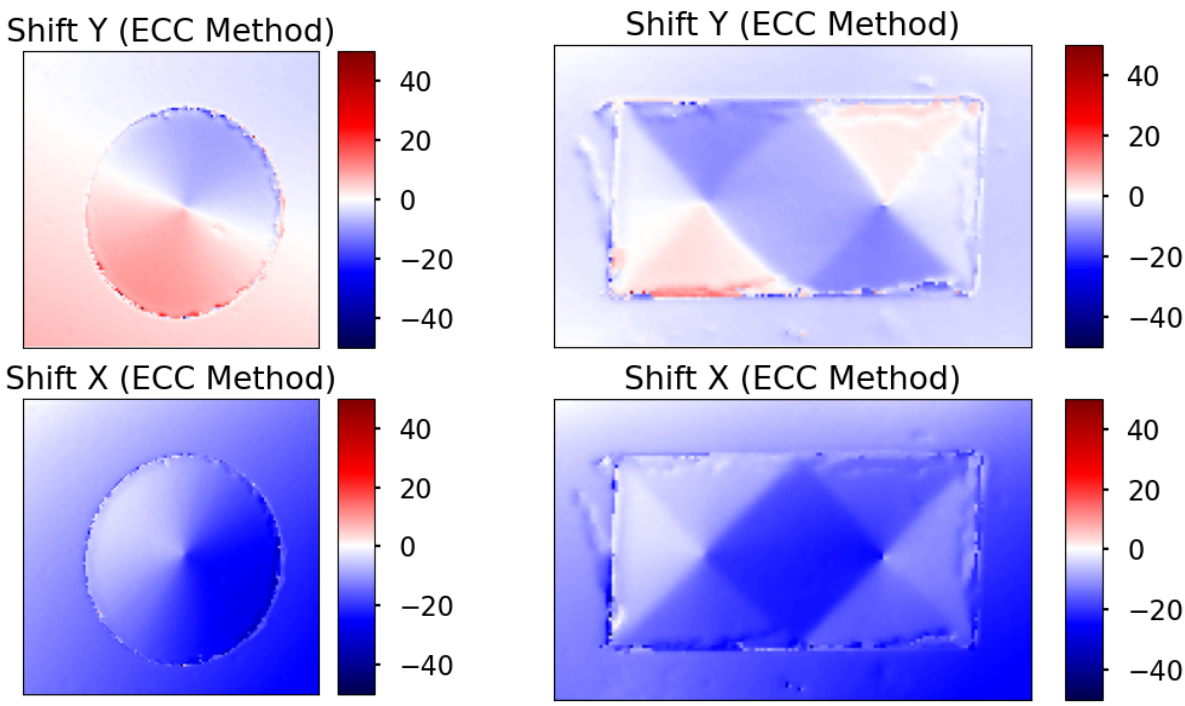

Figure 5: Beam shifts for the disk and rectangle calculated via the ECC method.

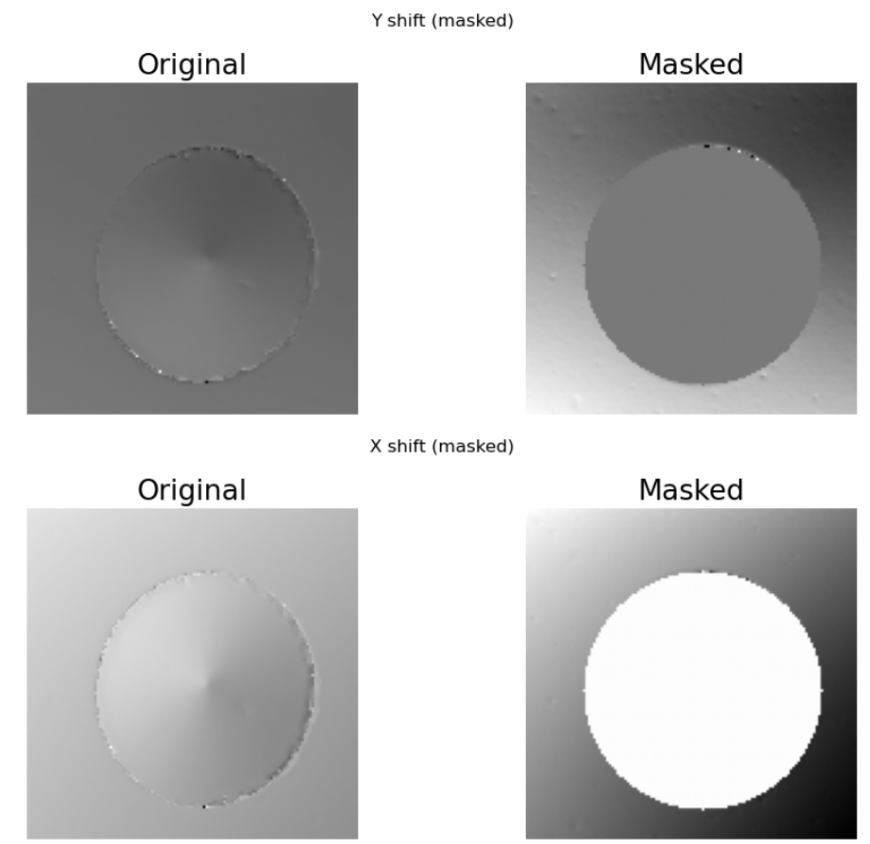

Figure 6: Example masks created for descan correction.

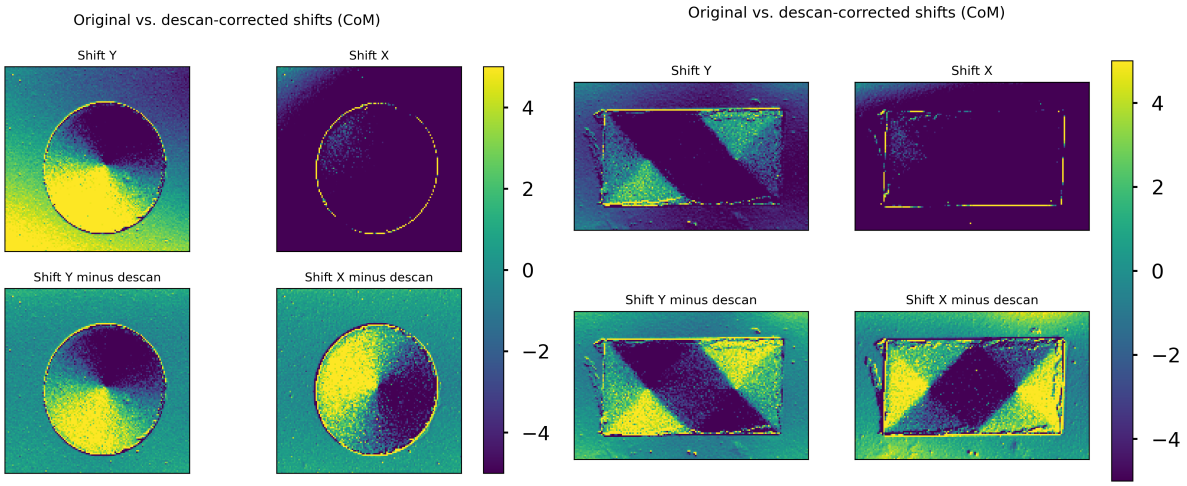

Figure 7: Descan-corrected shift maps for the CoM beam shifts.

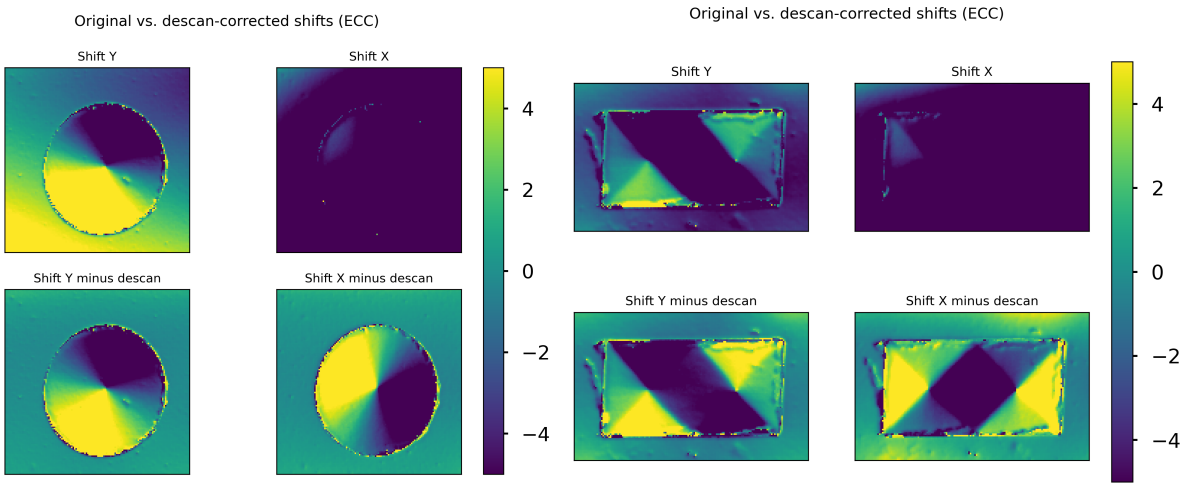

Figure 8: Descan-corrected shift maps for the ECC beam shifts.

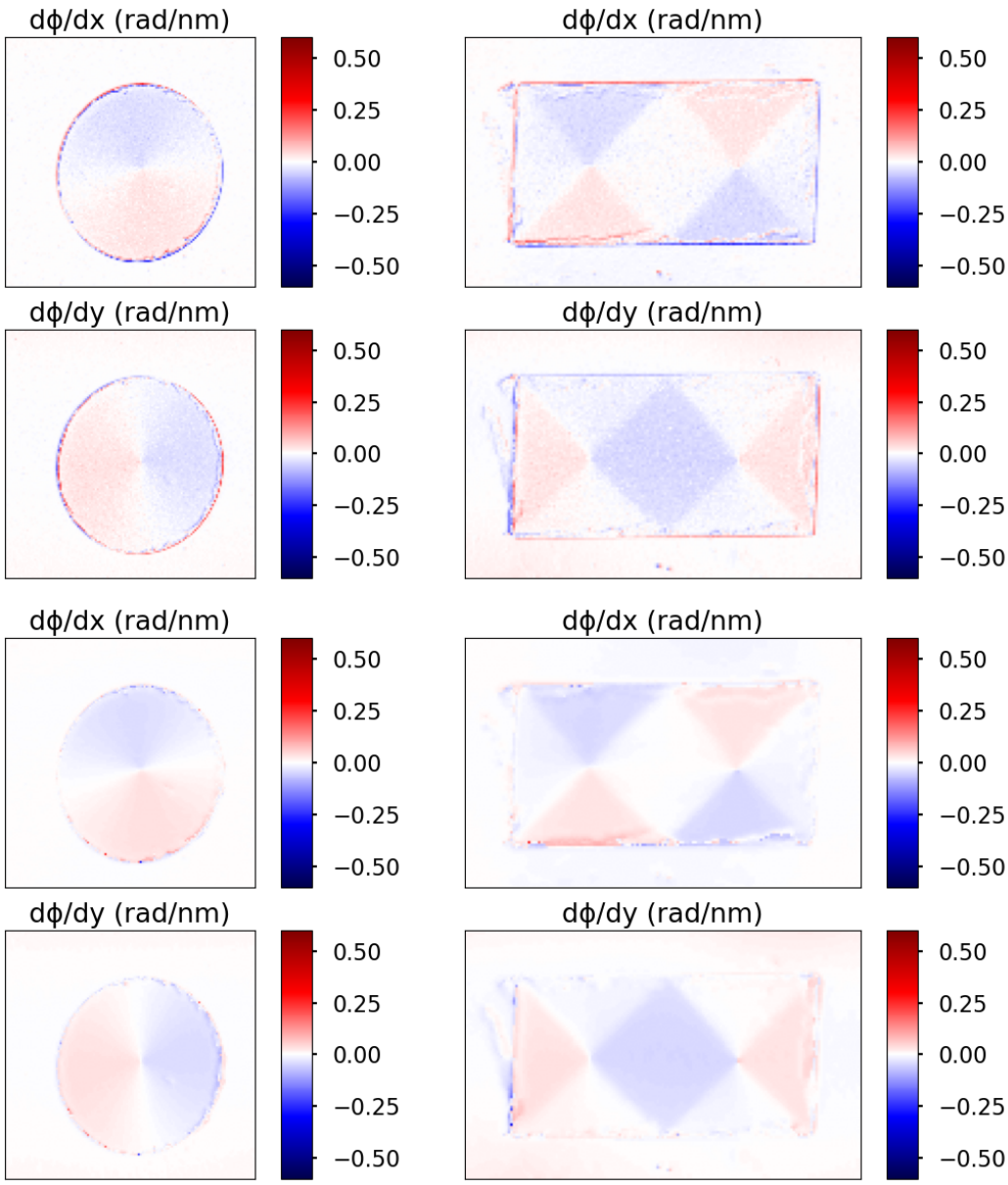

Figure 9: Phase gradients for CoM (top) and ECC (bottom).

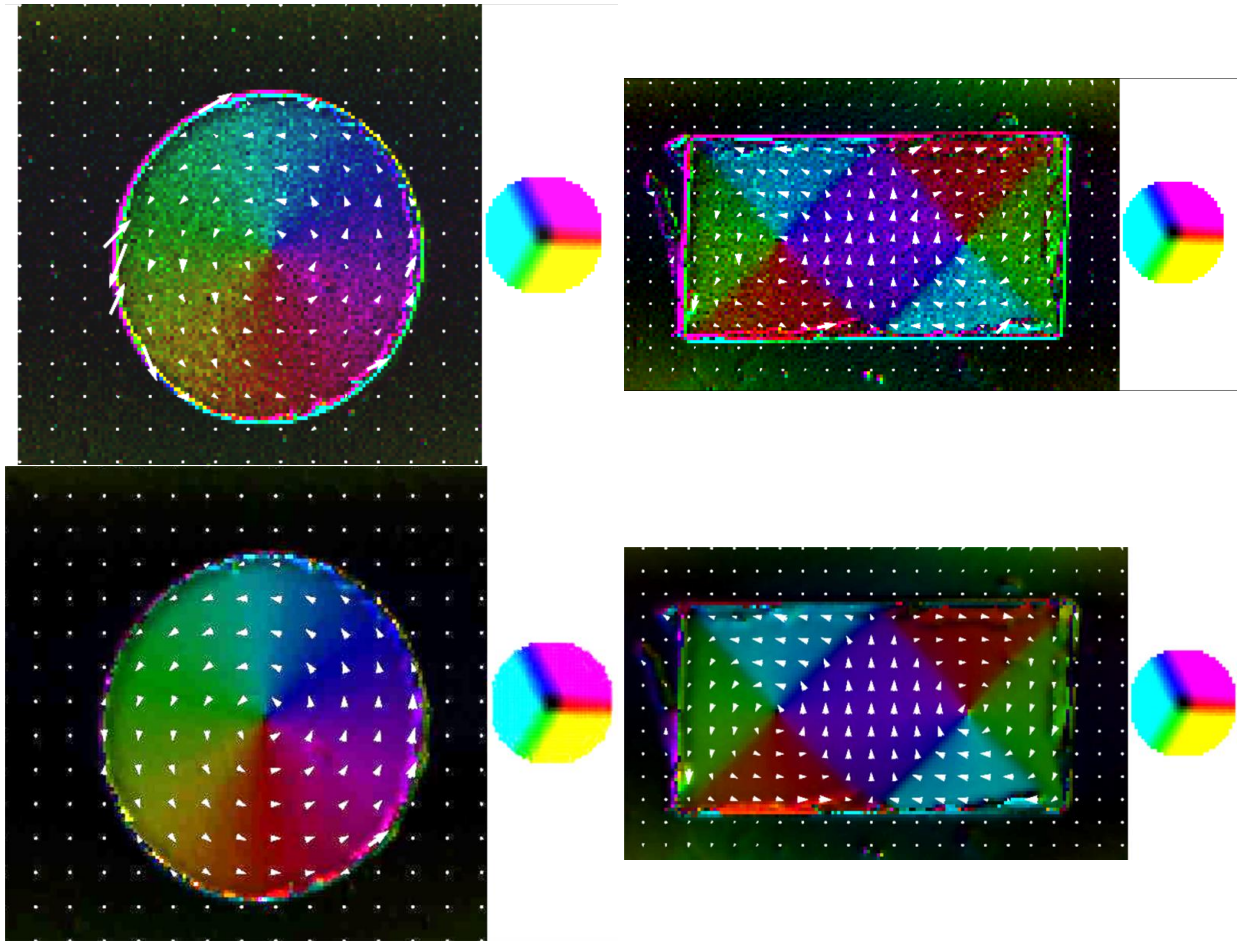

Figure 10: Final integrated induction maps for CoM (top) and ECC (bottom). The brightness and contrast have been adjusted (hence the non-uniform color map).

The magnetic field can be calculated from the phase gradient via the relationship

$$\boldsymbol{\nabla}_m \phi = -\frac{\pi t}{\phi_0}(\boldsymbol{B} \times \hat{\boldsymbol{n}})$$

where ϕ_0 is the magnetic flux quantum (Planck's constant divided by the charge of the electron), t is the thickness of the sample, and $\hat{\boldsymbol{n}}$ is a unit vector pointing along the beam direction [1]. Since this calculation works out to be just a scaling factor, the $\boldsymbol{B}$-field map qualitatively looks identical to Figure 9. However, one can combine the x- and y-shifts into one image to create an integrated induction map, as shown in Figure 10.

3.4 Analysis, interpretation, and comparison to literature

As mentioned in Section 2.1, there is a very solid theoretical [4][7] and experimental [8][9] understanding of the magnetic domain structure and behavior of ferromagnetic materials such as Permalloy. These kinds of materials minimize their total energy by eliminating stray fields that point out of the surface poles or sample edges ($\boldsymbol{M} \cdot \hat{\boldsymbol{n}} = 0$), resulting in the formation of vortex-like structures. Circular samples are favorable for this energy minimization, because the vortex is always parallel to the edges. However, the decreased symmetry in rectangular samples is less favorable, leading to the sample separating into discrete domains, a process called Landau patterning (shown in the right half of Figure 11). Domains facilitating

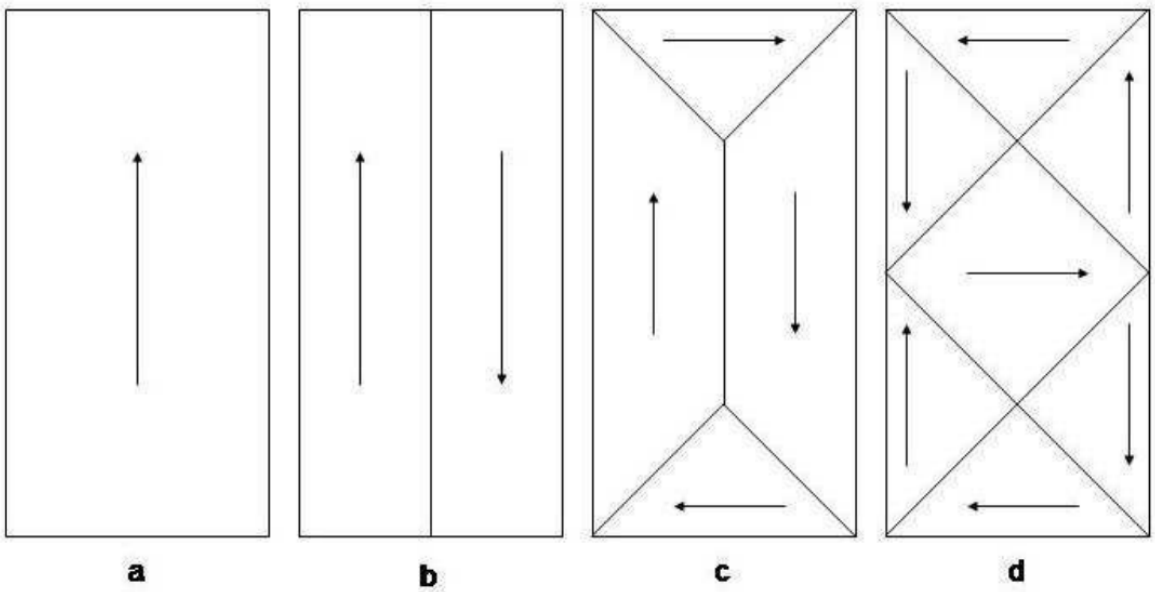

Figure 11: Schematic of domain formation, with Landau patterning occurring in (c) and (d). Reproduced from [4] courtesy of Y. de Boer (2005).

multiple vortices, as in Figure 11(d), can be formed depending on constraints such as the size and thickness of the sample, or the presence of defects. Permalloy, as a "soft" magnetic material, requires less energy to create domain walls, and thus can afford to further minimize energy by creating more of them.

The results obtained in this project agree with the expectation from the literature. With both the CoM and ECC methods, a counterclockwise vortex is obtained for the disk sample, and a Landau pattern with two opposite vortices is obtained for the rectangular sample — a structure exactly like Figure 11(d). Notably, the ECC result looks cleaner than the CoM one, with fewer edge effects, even after the latter was filtered. This indicates that there may have been some artifacts such as the ones described in Section 2.2. This is a relatively novel and useful result, since CoM is generally the more prevalent method, and an observed difference for a simple system like this one might translate to a significant improvement when imaging materials with more complex domain structures.

3.5 Limitations and future work

Ideally, the reference point for the beam shift calculations would be a vacuum probe — i.e. one would obtain beam images without actually imaging a sample to get a proper, unshifted background to subtract. Since this was not done when collecting the data, the top-left pixel (0,0) was used, which might still have had some intrinsic shift. Thus, the beam shifts would be improved both qualitatively and quantitatively with a proper vacuum probe. Furthermore, only a single median filter was used on the CoM data after the descan correction; more complex filtering techniques could be employed in the future to improve the quality of the final magnetic induction.

In this pipeline, the descan correction was calculated after the beam shifts, because the planar fit method must be done on real-space images. The pipeline could be improved by moving this step earlier in the process; that way, any descan-caused microscope error unaccounted for in the beam shift calculation would be resolved. A new method would have to be developed to calculate descan correction on diffraction images, simply because real-space data is not immediately extractable.

Aside from these improvements, one future extension to this project could be to add fields to the samples to ensure that the resulting changes in the magnetic domains would be accurately captured. This would be useful for characterizing samples with time-dependent dynamics.

4 Conclusion

This project concerned the use of DPC to detect magnetic fields in small circular and rectangular Permalloy ferromagnets. A pipeline was developed to calculate shifts of the diffraction beam (via two methods, center-of-mass and edge cross-correlation), phase gradients, and magnetic induction maps from raw 4D-STEM data. The results agree with the literature expectations for small samples, with a single vortex in the disk and Landau patterning with two vortices in the rectangle. Though further testing with added fields would be useful to ensure the pipeline's quality, it could be implemented on a variety of DPC data for magnetic samples. In particular, tracking the edges of the beam, though less frequently used than tracking the center, is shown here to yield slightly clearer maps, signaling a potential improvement of the technique.

5 Code and Data Availability

Code, datasets, and documentation are available in the project GitHub repository:

https://github.com/pravan007/MATSCI465_final_project/tree/main

This repository contains:

- `ECC_DiskTracking.ipynb` and `CoM_DiskTracking.ipynb` — Notebooks for calculating beam shifts from raw .mrc files
- `PhaseBCalculations.ipynb` — Notebook for calculating phase gradients and magnetic induction from beam shift .npy files
- Folder `dpc4dstem` containing necessary Python scripts for running the above notebooks
- Input and output folders for the relevant data/results [Note: the .mrc files were not included due to file size constraints. They can be accessed through the following Box folders: (disk) (rectangle)]
- Partial list of references consulted
- This paper
- In-class presentation slides
- README and environment file

6 Acknowledgments

The author would like to thank Drs. Yue Li and Daniel Durham for their help and guidance on the conceptual, computational, and experimental challenges of this project. The author would also like to thank the creators of PyLorentz [5], as well as his advisor, Prof. Charudatta Phatak. Lastly, the author thanks Prof. Roberto dos Reis for his support and guidance throughout the quarter.

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